thermoscientific

2020 Training Programme

Stable Isotope Ratio Mass Spectrometry

and

Isotope Ratio Infrared Spectrometry

Table of Contents

1. General Information 3

2. Visa Formalities 4

3. Training Arrangements

   3.1 EA-IRMS 3-day Training Course
   Flash for IRMS/EA IsoLink/TC-EA; ConFlo IV 6

   3.2 GC-IRMS 4-day Training Course
   GC IsoLink and GC Peripherals 7

   3.3 LC-IRMS 4-day Training Course
   LC IsoLink and LC Peripherals 8

   3.4 GB-IRMS 4-day Training Course
   GasBench II 9

   3.5 KIEL IV-Dual Inlet -IRMS 4-day Training Course:
   KIEL IV Carbonate Device and Peripherals 10

   3.6 Dual Inlet-IRMS 3-day Onsite Training Course:
   Dual Inlet and Peripherals 11

4. Isotope Ratio Infrared Spectrometry (IRIS)
   Delta Ray 3-day Training Course 12

5. 2020 Training Registration Form 13

6. 2020 Trainings Schedule 14

1. General Information

Thermo Fisher Scientific (Bremen) GmbH is pleased to offer training courses throughout 2020 for stable isotope ratio mass spectrometry and isotope ratio infrared spectrometry, associated peripherals and software. We offer a comprehensive, certified training programme focussed on your peripherals and isotope ratio mass spectrometer, led by our experienced experts in Bremen that will support you in maximising the throughput and quality of output from your instrumentation. Our aim is to provide you with an overview of:

- the mass spectrometry and peripheral hardware and their functionality
- the operation of the software, sample analysis and data evaluation
- basic maintenance
- the Isotope Ratio Infrared Spectrometry (IRIS) operation of the software, sample analysis and data evaluation

Our training courses take place in our Application Laboratory at the Thermo Fisher Scientific, Bremen, Germany factory but we also offer on-site training, meaning our specialists will come to your laboratory. These on-site training courses will be customized according to your instrument(s), software and application requirement(s) and your own instrument(s) will be used.

Each course requires a minimum of three participants. The training course language will be English unless specified otherwise.

For **training fees** please contact your local Thermo Fisher Scientific sales representative. The training fees include catering during training session (two coffee breaks and lunch). The training fees do not include individual travel costs, hotel accommodation or per diem expenses.

We offer to assist in booking rooms at the Holiday Inn Express Bremen Airport Hotel, which is conveniently located within walking distance of the Bremen Airport and right next to our facility, approx. 15 minutes by tram from the main railway station. The rate is 81 € including breakfast, Hi-speed and internet access and WLAN.

Registration deadline is *eight weeks* prior to scheduled dates, after which time you may only register for later courses. For registration please use the registration form located at the back of this brochure.

In case of cancellation we strongly recommend to nominate a substitute if possible. Cancellation prior to registration deadline is possible. Cancellations two weeks prior to scheduled courses are subject to a 50 % cancellation fee. Cancellations a week prior to scheduled courses are subject to a 90 % cancellation fee.

Training courses may be rescheduled or cancelled due to a lack of participants or other reasons. Already registered attendees will be informed as soon as possible.

2. Visa Formalities

If you require an entry visa for Germany we ask that you complete this form. This will allow Thermo Fisher Scientific (Bremen) GmbH to issue an invitation letter to support your application for a visa to enter Germany. Please mail this form to susanne.tobin@thermofisher.com or fax it to our office at ++49 (421) 54 93 396.

Please allow 2-3 weeks for processing and mailing of the invitation letter.

Last name (surname)	
First name(s) **(given name(s))**	
Title	
Gender (male / female)	
Date of birth DD - month (in letters) - YYYY	
Passport number & expiry date	
Country of Citizenship	
Dates of planned entry into and exit from Germany	
Correct and complete name and address of customer's company or institute, including telephone and fax numbers (no postbox):	
Expenses for travel, lodging, health insurance and per diem will be borne by:	
Correct and complete address of the embassy where to apply for visa, including telephone and fax numbers:	
Correct and complete address (no post box) of where the originals of the invitation letters shall be sent by courier, including contact person, telephone and fax numbers:	

3. Training Arrangements

The Thermo Fisher Scientific Isotope Ratio Mass Spectrometer training courses are designed for customers who have purchased the Thermo Scientific DELTA and 253 PLUS Plus series mass spectrometers and the given interfaces and peripherals or a Thermo Scientific Delta Ray Isotope Ratio Infrared Spectrometer. The **instruments used for training** will be DELTA V with the **software** Isodat 3.0 or a Delta Ray with Qtegra Software.
Our training courses are structured as **3-day or 4-day courses** and designed to familiarize you with your instruments, covering basic operation and maintenance, through software control, data analysis and evaluation and troubleshooting, covering the general aspects of isotope ratio mass spectrometry (253 PLUS or DELTA V) for 1 and a half days with the remaining time dedicated to the peripheral.
Please select your course carefully based on the instruments that you have and if necessary please contact us to discuss so that we can serve your needs best.

Participants' profile: Basic understanding of mass spectrometry (MS); Proficiency in English language; 3 – 6 months experience with Thermo Fisher Scientific IRMS instrument and peripherals.

Training hours

Day 1	10 – 17:00 h
Day 2	9 – 16:30 h
Day 3	9 – 16:30 h
Day 4	9 – 15:30 h

Training courses	Training Dates
EA-IRMS: Flash for IRMS/EA IsoLink/TC-EA and ConFlo IV *3-day* training course Part number: 0736010	**February 19-21, 2020** **November 18-20, 2020**
GasBench II IRMS Operator Training 4-day course Part number 0736030	**March 01-04, 2020** **November 09-12, 2020**
GC-IRMS: GC IsoLink and GC peripherals *4-day* training course Part number 0736020	**February 25-28, 2020** **April 21-24, 2020** **November 23-26, 2020**
KIEL IV-Dual Inlet IRMS: Kiel IV Carbonate Device *4-day* training course Part number 0736040	**November 01-04, 2020**

3.1 EA – IRMS 3-day Training Course

Part number: 0736010

The EA – IRMS course is designed for users interested in the analysis of bulk materials for stable isotopes of carbon, nitrogen and sulphur by combustion and oxygen and hydrogen by pyrolysis. The course is specific to the Thermo Scientific Flash series elemental analysers for IRMS, TC/EA, EA IsoLink, Con Flo IV interface device and the Delta V series mass spectrometers. The instruction will include the theory of IRMS, from ionisation through ion detection and production of isotope ratios, the interface between the IRMS and elemental analyser(s) and the theory behind elemental analysers. In addition, we cover the Isodat software platform, including developing methods and sequences and data evaluation and data correction strategies. Further, there is dedicated time for hands-on with the instrument to apply the course material for real!

The EA – IRMS course material includes:

Isotope Ratio Mass Spectrometry

IRMS Theory

- Ionisation and ion source tuning
- Mass analyzer and electromagnet
- Ion detection & amplifier
- Vacuum system

IRMS basic maintenance

- Ion source mounting, demounting and cleaning
- Pump maintenance
- Spare parts

Software

Isodat 3.0 software platform

- File structure and data flow
- IRMS Mass calibration
- Method and sequence development
- Data acquisition
- Post- analysis data evaluation
- Data import and export

ConFlo IV interface

ConFlo IV

- Principle of ConFlo IV
- Helium dilution strategies
- Reference gases
- Linearity and reproducibility checks for IRMS

Elemental Analyzers (CNSOH)

- Overview of Elemental Analysis
- Reactors and maintenance
- Sample preparation and handling
- Measurement strategies
- Data corrections and normalisation
- Hands-on EA – IRMS analysis, covering sequence production through to data evaluation and correction

This course is offered at the factory in Bremen or at the customer site.

3.2 GC-IRMS 4-day Training Course

Part number 0736020

The **general objective** of this **4-day** operator training course is to familiarize the operator with DELTA V Plus and DELTA V Advantage instrument features, vacuum system, ion source setting and cleaning, basic instrument operations, instrument check and maintenance basics, software and troubleshooting. It intends to give you the best understanding for the GC IsoLink, and GC peripherals.

Day 1

Introduction

Basic Theory
- Ion source operation
- Analyzer and electromagnet
- Ion detection & amplifier

Vacuum System
- General introduction

Maintenance
- Ion source mounting, demounting and cleaning
- Pump maintenance
- Spare parts

Application Software
- Instrument control module
- Mass calibration
- System and signal stability
- File structure and data flow
- Result workshop
- Data import and export

Day 2-4

GC Interface and Peripherals

GC IsoLink
- General introduction
- Gas flow modes
- Interface components
- Tests after GC IsoLink maintenance

GC Combustion (GC/C)
- Reactors and conditioning
- C, N sample handling
- Isotope ratio determination and analytical approaches

GC High Temperature Conversion (GC/TC)
- Reactors and conditioning
- H, O sample handling
- Isotope ratio determination and analytical approaches

GC-IRMS
- Basic Requirements on GC
- Injection methods
- GC columns
- Optimizing of GC pathways
- GC data evaluation
- Autosampler methods

Maintenance

Data Processing, Corrections, Calibration, Referencing

This course is offered at the factory in Bremen or at the customer site.

3.3 LC-IRMS 4-day Training Course

Part number 0736060

The **general objective** of this **4-day** operator training course is to familiarize the operator with DELTA V Plus and DELTA V Advantage instrument features, vacuum system, ion source setting, basic instrument operations, instrument check and maintenance basics, software and troubleshooting. It intends to give you the best understanding for the interfaces and sample preparation units LC IsoLink and LC peripherals.

Day 1

Introduction

Basic Theory
- Ion source operation
- Analyzer and electromagnet
- Ion detection & amplifier

Vacuum System
- General introduction

Maintenance
- Ion source mounting, demounting and cleaning
- Pump maintenance
- Spare parts

Application Software
- Instrument control module
- Mass calibration
- System and signal stability
- File structure and data flow
- Result workshop
- Data import and export

Day 2-4

LC IsoLink Interface

Basic Operations
- Introduction in LC-IRMS
- Setup of the LC IsoLink
- Components of the LC IsoLink
- How to control the LC IsoLink and HPLC system with ISODAT
- Operating modes:
  - Bulk injection mode (µ-EA mode)
  - HPLC mode (CSIA mode)

Application
- Getting started
- Liquid flow area of the interface
- Gas isotope area of the interface
- Requirements on the HPLC systems
- How to start with a new application
- Method Development
- Application examples
- Maintenance of the components
- Dionex UltiMate 3000 LC Pump
- Dionex UltiMate 3000 LC Autosampler

Data Processing, Corrections, Calibration, Referencing

This course is offered at the factory in Bremen or at the customer site.

3.4 GB-IRMS 4- day Training Course

Part number 0736030

The **general objective** of this **4-day** operator training course is to familiarize the operator with DELTA V Plus and DELTA V Advantage instrument features, vacuum system, ion source setting and cleaning, basic instrument operations, instrument check and maintenance basics, software and troubleshooting. It intends to give you the best understanding for the GasBench II.

Day 1

Introduction

Basic Theory
- Ion source operation
- Analyzer and electromagnet
- Ion detection & amplifier

Vacuum System
- General introduction

Maintenance
- Ion source mounting, demounting and cleaning
- Pump maintenance
- Spare parts

Application Software
- Instrument control module
- Mass calibration
- System and signal stability
- File structure and data flow
- Result workshop
- Data import and export

Day 2-4

GasBench II

Basic Operations
- VALCO 8 port valve and switching sequence
- Nafion water traps
- Tray and GC column temperature setting
- Open split and reference inlet
- Carbonate option
- Autosampler operation
- Isotope ratio determination

Applications
- CO_2 in air
- Breath tests
- Water equilibration for ^{18}O and HD
- Carbonates
- Dissolved inorganic carbon
- Atmospheric gases

Data Processing, Corrections, Calibration, Referencing

This course is offered at the factory in Bremen or at the customer site.

3.5 Kiel IV-Dual Inlet 4- day Training Course

Part number 0736040

The **general objective** of this **4-day** operator training course is to familiarize the operator with DELTA V Plus and DELTA V Advantage instrument features, vacuum system, ion source setting and cleaning, basic instrument operations, instrument check and maintenance basics, software and troubleshooting. It intends to give you the best understanding for the KIEL IV Carbonate Device.

Day 1

Introduction

Basic Theory
- Ion source operation
- Analyzer and electromagnet
- Ion detection & amplifier

Vacuum System
- General introduction

Maintenance
- Ion source mounting, demounting and cleaning
- Pump maintenance
- Spare parts

Application Software
- Instrument control module
- Mass calibration
- System and signal stability
- File structure and data flow
- Result workshop
- Data import and export

Day 2

Dual Inlet and Peripherals

Dual Inlet
- Principle of operation, bellows and valves
- Pressure adjust
- Sample CO_2 acquisition
- Dual inlet system
- Isotope ratio determination (normalization & standardisation)
- Process control and evaluation
- Maintenance
- Microvolume
- Multiport

Day 3-4

KIEL IV Carbonate Device

KIEL IV
- General introduction
- All Interface components
- Tests after maintenance
- Leak testing modes
- Sample handling
- Isotope ratio determination and analytical approaches
- Clumped-CO2 of carbonates with Kiel IV
- Data evaluation
- Special troubleshooting
- Maintenance

Data Processing, Corrections, Calibration, Referencing

This course is offered at the factory in Bremen or at the customer site.

3.6 Dual Inlet–IRMS 3-day onsite Training Course

Part number 0736050

The **general objective** of this **3-day** operator training course is to familiarize the operator with the DELTA V Plus and DELTA V Advantage instrument features, vacuum system, ion source setting, basic instrument operations, instrument check and maintenance basics, software and trouble shooting. It intends to give you the best understanding for Dual Inlet operation with peripherals including Multiport and Microvolume.

Day 1

Introduction

Basic Theory
- Ion source operation
- Analyzer and electromagnet
- Ion detection & amplifier

Vacuum System
- General introduction

Maintenance
- Ion source mounting, demounting and cleaning
- Pump maintenance
- Spare parts

Application Software
- Instrument control module
- Mass calibration
- System and signal stability
- File structure and data flow
- Result workshop
- Data import and export

Day 2-3

Dual Inlet and Peripherals

Dual Inlet
- Principle of operation, bellows and valves
- Pressure adjust
- Sample CO_2 acquisition
- Dual inlet system
- Isotope ratio determination (normalization & standardisation)
- Process control and evaluation
- Maintenance
- Microvolume
- Multiport

Data Processing, Corrections, Calibration, Referencing

This course is offered at the customer site only.

4. IRIS Isotope Ratio Infrared Spectrometry 3- day Training Course

Part number 1358390

The **3-day** IRIS operator training course is designed for new users of the Delta Ray instrument. The instruction will include the theory of IRIS, from absorption spectra to production of isotope ratios, the Universal Reference Interface for advanced calibration and referencing schemes as well as the Xpand Dilution kit to analyse samples with up to 100% CO2 concentration. In addition, we cover the Qtegra software platform, including general settings and methods settings, especially the advanced settings, data evaluation and data export and reporting functions. Further, troubleshooting strategies are presented and there is dedicated time for hands-on with the instrument to practice on real samples and real applications!

IRIS Theory

- Different IRIS approaches
- Direct absorption spectrometer
- Vacuum system

Software

Qtegra software

- Templates
- Labbooks
- General and Method settings
- Advanced calibration settings
- Data acquisition
- Post- analysis data evaluation
- Data import and exp

Universal Reference interface

- Sample Drying
- Calibration gases
- Sample dilution

Xpand

- Principle of Xpand
- Dynamic Dilution Mode
- Fixed Dilution Modes
- How to select the dilution mode
- How to sample from a vial
- How to sample from a continuous high concentration source

This course is offered at the factory in Bremen or at the customer site.

5. Customer Operator Training 2020 Registration Form

Please mail this form to susanne.tobin@thermofisher.com or fax it to our office at ++49 (421) 54 93 396. Thank you.

Participant	Training Course / Instruments:
Last name: First name: Title: Company/Institute: Address (street): Postal code: City / State: Country: Phone: Fax: E-Mail:	Date: **Accommodation** Hotel reservation Yes ☐ No ☐ Smoking ☐ Non-smoking ☐ At Holiday Inn Express Bremen Airport ☐ 81 € including breakfast and internet access) Other ☐ (please specify price range / location) Date of Arrival: Date of Departure:
What instruments do you have??? _____ Service number (if known): S/N _____ **We use the following software** _____ **Our main application is** _____	**Please enter the Sales Order Number here:** _____ **Remarks:**

Date _____ Signature _____

6. Customer Training Course Overview

Instruments	Training Course
Isotope Ratio Mass Spectrometers **DELTA Series** **253 PLUS / MAT 253** **Isotope Ratio Infrared Spectrometer**	**EA-IRMS** Elemental Analyzer and TC/EA Peripherals 3-day course **GC-IRMS** GC IsoLink and GC peripherals 4-day course **LC-IRMS** LC IsoLink and LC peripherals 4-day course **GB-IRMS** GasBench II and GB peripherals 4-day course **KIEL IV-Dual Inlet IRMS** Kiel IV Carbonate Device 4-day course **Dual Inlet**: Dual Inlet and Peripherals 3-day course (onsite only) **Delta Ray** 3-day course
Thermal Ionization Mass Spectrometer **TRITON** **TRITON *Plus***	TRITON and TRITON *Plus* Basic Operator Training 5-day course Instrument Maintenance Training 2-day course, *Upon request*
Multicollector ICP-MS **NEPTUNE** **NEPTUNE *Plus***	NEPTUNE and NEPTUNE *Plus* Basic Operator Training 5-day course Instrument Maintenance Training 2-day course, *Upon request*
ICP-MS / GD-MS **ELEMENT** **ELEMENT 2** **ELEMENT XR** **ELEMENT GD Plus**	ELEMENT2 / ELEMENT XR Operator Training 5-day course ELEMENT GD Plus Operator Training 5-day course
DFS	DFS Instrument Operator Training for Quantitative Applications 5-day course DFS Instrument Operator Training for Qualitative Applications and Characterization, *Upon request* DFS Dual Data Operator Training, *Upon request*